Vectors in n Dimensions

Vector addition and scalar multiplication

1 Let $\mathbf{u} = \langle 2, -1, 3\rangle$, $\mathbf{v} = \langle 1, 4, -2\rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.

Dot product and norm

2 For $\mathbf{u} = \langle 1, 2, 2\rangle$, $\mathbf{v} = \langle 3, -1, 0\rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.

3 Find the angle between $\mathbf{u} = \langle 1, 0, 0\rangle$ and $\mathbf{v} = \langle 1, 1, 0\rangle$ using $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

The Cauchy–Schwarz Inequality

4 State the Cauchy–Schwarz Inequality.

5 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4\rangle$, $\mathbf{v} = \langle 1, 2\rangle$.

Unit vectors

6 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4\rangle$.

Orthogonality

7 Determine whether $\mathbf{u} = \langle 2, -3, 1\rangle$ and $\mathbf{v} = \langle 1, 2, 4\rangle$ are orthogonal.

8 For $\mathbf{u} = \langle 1, -1, 2\rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.